

# A Trigger Construction for Reversed Strategy–Control Equilibria

## Rigorous Partial Progress for a Common-Noise Dividend Game

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**Status.** This paper gives *partial progress, not a complete solution* of the reversed-role equilibrium problem posed by De Angelis, Gensbittel, and Villeneuve. For every  $0 < x < y$  it constructs an explicit pure profile in which the poorer firm uses a strategy and the richer firm uses a control. The profile is a Nash equilibrium up to an error whose *exact* value is identified as a one-player punishment premium and whose size is bounded explicitly. The poorer firm is an exact best response. The error tends to zero near the diagonal and, uniformly in the richer firm’s lead, when the poorer reserve is large. The remaining exactness question is reduced to a concrete one-player singular-control value  $W(x, y)$ .

### Abstract

We study the open question of whether, in the common-noise competitive dividend model of De Angelis, Gensbittel, and Villeneuve, there is a Nash equilibrium in which the poorer firm uses a nonanticipative strategy while the richer firm uses an opponent-independent control. A previously proposed compactness proof cannot be repaired locally: it tests the wrong deviation class for randomized profiles, loses responses on off-equilibrium paths, and relies on a false payoff-density assertion at time-zero jumps.

We replace that argument by an explicit pure trigger construction. The richer firm pays a time-zero concession, reducing its reserve to  $z_* := \min\{x, a_0\}$ , and then uses the classical dividend control. If and only if that concession is observed, the poorer firm switches to the predatory pure strategy from the known asymmetric (or equal-reserve asymmetric) equilibrium; otherwise it withholds all dividends. The poorer firm is an exact best response. The richer firm’s exact deviation gain is

$$\varepsilon_*(x, y) = W(x, y) - (y - z_* + v_0(z_*)),$$

where  $W(x, y)$  is its optimal payoff against a rival that never pays dividends. We prove

$$0 \leq \varepsilon_*(x, y) \leq v_0(y) - v_0(z_*) - (y - z_*) + e^{-\kappa x} \max_{0 \leq s \leq y-x} (\hat{v}(s) - v_0(s)),$$

with  $\kappa = (\mu_0 + \sqrt{\mu_0^2 + 2r\sigma^2})/\sigma^2$ . Because the constructed profile is pure, the estimate holds against the full randomized-strategy deviation class of the original game. This yields rigorous approximate equilibria, a necessary security-value condition for every exact reversed equilibrium, and a sharply isolated remaining problem; it does not settle exact existence for all states and parameters.

## 1 Model and the imported equilibrium theorem

We use the model and strategy spaces of De Angelis, Gensbittel, and Villeneuve [1]. The brief description below fixes notation and records precisely the part of their analysis used in our construction.

Fix parameters

$$r > 0, \quad \sigma > 0, \quad 0 < \mu_0 < \hat{\mu},$$

and let  $B$  be a Brownian motion. Before the first default, the reserves of firms with initial cash  $x$  and  $y$  are

$$X_t^L = x + \mu_0 t + \sigma B_t - L_t, \quad Y_t^D = y + \mu_0 t + \sigma B_t - D_t. \quad (1)$$

Here  $L$  and  $D$  are right-continuous, nondecreasing cumulative dividend controls, with possible jumps at time zero, and with no jump exceeding the available reserve. Write

$$\tau_X(L) := \inf\{t \geq 0 : X_t^L \leq 0\}, \quad \tau_Y(D) := \inf\{t \geq 0 : Y_t^D \leq 0\}.$$

If one firm defaults strictly before the other, the survivor becomes a monopolist. If  $\hat{v}$  denotes the classical monopoly dividend value with drift  $\hat{\mu}$ , the second firm's payoff is

$$J_2^{x,y}(D, L) := \mathbb{E} \left[ \int_{[0, \tau_X(L) \wedge \tau_Y(D)]} e^{-rt} dD_t + \mathbf{1}_{\{\tau_X(L) < \tau_Y(D)\}} e^{-r\tau_X(L)} \hat{v}(Y_{\tau_X(L)}^D) \right], \quad (2)$$

and  $J_1^{x,y}$  is defined symmetrically.

A pure control is a nonanticipative functional of the Brownian path alone. A pure strategy may additionally react nonanticipatively to the opponent's cumulative dividend path. Randomized strategies add an independent private randomizer. We use  $\mathcal{D}(a)$  for pure controls from reserve  $a$  and  $\mathcal{S}(a)$  for pure strategies. A strategy-control pair is triangular and therefore induces a unique pair of controls. The full game nevertheless permits unilateral deviations to arbitrary randomized strategies; a non-control-inducing deviation receives payoff  $-\infty$ , as in [1].

For a drift  $\mu > 0$ , let  $w(\cdot; \mu)$  be the value of the classical one-firm dividend problem and  $a^*(\mu)$  its reflecting barrier. Set

$$v_0 := w(\cdot; \mu_0), \quad \hat{v} := w(\cdot; \hat{\mu}), \quad a_0 := a^*(\mu_0), \quad \hat{a} := a^*(\hat{\mu}).$$

The standard verification theorem gives

$$v_0(0) = \hat{v}(0) = 0, \quad v'_0 \geq 1, \quad \hat{v}' \geq 1, \quad v'_0(s) > 1 \iff 0 < s < a_0, \quad (3)$$

and both value functions are affine with slope one above their respective barriers. Moreover  $\hat{v} \geq v_0$ , because every policy admissible under drift  $\mu_0$  remains admissible under the larger drift  $\hat{\mu}$ .

For  $p > 0$ , denote the classical control from reserve  $p$  by

$$C_t^p := \sup_{0 \leq s \leq t} (p - a_0 + \mu_0 s + \sigma B_s)^+. \quad (4)$$

In particular,  $C_0^p = 0$  when  $p \leq a_0$ .

We now isolate the theorem imported from [1, Theorem 4.1 and Remark 4.10].

**Theorem 1.1** (Known asymmetric equilibrium). *For every  $0 < p < q$  there is a pure strategy  $G^{q;p} \in \mathcal{S}(q)$  such that, in the game started from reserves  $(q, p)$ , the profile*

$$(G^{q;p}, C^p)$$

*with the first player designated as the richer strategic player and the second as the poorer control player is a Nash equilibrium of the full randomized-strategy game. The control player's payoff is  $v_0(p)$ . The same statement holds for  $p = q$  after choosing one of the two asymmetric pure equilibria of [1, Remark 4.10]; the designated control player again receives  $v_0(p)$ .*

*Equivalently, for every pure control  $A \in \mathcal{D}(q)$  and every pure control  $E \in \mathcal{D}(p)$ ,*

$$J_{\text{strat}}^{q,p}(G^{q;p}, C^p) \geq J_{\text{strat}}^{q,p}(A, C^p), \quad (5)$$

$$v_0(p) \geq J_{\text{ctrl}}^{q,p}(E, G^{q;p}[E]). \quad (6)$$

The notation  $G^{q;p}[E]$  means the control induced when the strategic player observes the control  $E$ . We shall use theorem 1.1 as a black box; all new arguments below are independent of the free-boundary calculations used to prove it.

## 2 Two structural lemmas

The first lemma is the approximate counterpart of [1, Lemma 3.7(iii)]. It is important because it permits us to work with pure profiles while retaining the original game's full randomized deviation class.

**Lemma 2.1** (Pure-control test with errors). *Let  $(S_1, S_2) \in \mathcal{S}(x) \times \mathcal{S}(y)$  be a control-inducing pure profile. Suppose that numbers  $\varepsilon_1, \varepsilon_2 \geq 0$  satisfy*

$$\begin{aligned} J_1^{x,y}(A, S_2[A]) &\leq J_1^{x,y}(S_1, S_2) + \varepsilon_1 && \text{for every } A \in \mathcal{D}(x), \\ J_2^{x,y}(E, S_1[E]) &\leq J_2^{x,y}(S_2, S_1) + \varepsilon_2 && \text{for every } E \in \mathcal{D}(y). \end{aligned}$$

*Then  $(S_1, S_2)$  is an  $(\varepsilon_1, \varepsilon_2)$ -Nash equilibrium against arbitrary randomized strategy deviations.*

*Proof.* Consider a randomized deviation by Player 1. If the deviating pair is not control-inducing, its payoff is  $-\infty$  and there is nothing to prove. Otherwise, because  $S_2$  is pure, conditioning on Player 1's private randomizer  $u$  produces for almost every  $u$  a pure control  $A^u \in \mathcal{D}(x)$  and its uniquely induced opponent control  $S_2[A^u]$ . Thus, using the payoff definition for randomized strategies,

$$J_1^{x,y}(\text{deviation}, S_2) = \int_0^1 J_1^{x,y}(A^u, S_2[A^u]) \, du \leq J_1^{x,y}(S_1, S_2) + \varepsilon_1.$$

The argument for Player 2 is identical. This is exactly the proof of [1, Lemma 3.7(iii)], with an additive error retained throughout.  $\square$

Our second structural lemma identifies the harshest possible response to a fixed control of Player 2.

**Lemma 2.2** (Withholding is the harshest response). *Fix  $D \in \mathcal{D}(y)$ . Let  $L^-, L^+ \in \mathcal{D}(x)$  satisfy  $L_t^- \leq L_t^+$  for every  $t$  and every Brownian path. Then*

$$J_2^{x,y}(D, L^-) \leq J_2^{x,y}(D, L^+). \quad (7)$$

*In particular,*

$$J_2^{x,y}(D, 0) \leq J_2^{x,y}(D, L) \quad \text{for every } L \in \mathcal{D}(x). \quad (8)$$

*The same conclusion holds after averaging over a randomized response  $L$ .*

*Proof.* The order of the dividend paths implies  $X^{L^+} \leq X^{L^-}$ , hence  $\tau_X(L^+) \leq \tau_X(L^-)$ . Up to  $\tau_X(L^+) \wedge \tau_Y(D)$  the two payoffs coincide. On the event  $\{\tau_X(L^+) < \tau_Y(D)\}$ , compare the continuation under  $L^-$  with the immediate monopoly value under  $L^+$ . The process

$$e^{-rt} \widehat{v}(Y_t^D) + \int_{[0,t]} e^{-rs} \, dD_s \quad (9)$$

is a supermartingale after localization, up to  $\tau_Y(D)$ . Indeed, in the continuous part,

$$\left( \frac{\sigma^2}{2} \partial_{yy} + \mu_0 \partial_y - r \right) \widehat{v} = \left( \frac{\sigma^2}{2} \partial_{yy} + \widehat{\mu} \partial_y - r \right) \widehat{v} - (\widehat{\mu} - \mu_0) \widehat{v}' \leq 0,$$

while a dividend jump of size  $d$  changes the sum in (9) by  $\hat{v}(Y_- - d) - \hat{v}(Y_-) + d \leq 0$  because  $\hat{v}' \geq 1$ . Optional sampling from  $\tau_X(L^+)$  to  $\tau_X(L^-) \wedge \tau_Y(D)$  therefore shows that the expected dividends paid during that interval, plus the possible later monopoly value, do not exceed  $\hat{v}(Y_{\tau_X(L^+)}^D)$ . This is exactly (7) after conditioning on  $\mathcal{F}_{\tau_X(L^+)}$ . Standard localization and the linear growth of  $\hat{v}$  justify removal of the localizing sequence. Since every cumulative dividend path is nonnegative, taking  $L^- = 0$  gives (8). Pointwise application followed by integration gives the randomized statement.  $\square$

**Definition 2.3** (Punishment value). *For  $0 < x < y$ , define*

$$W(x, y) := \sup_{D \in \mathcal{D}(y)} J_2^{x, y}(D, 0). \quad (10)$$

*Thus  $W(x, y)$  is the richer firm's optimal payoff when the poorer rival withholds all dividends.*

**Corollary 2.4** (Security value). *Against any pure or randomized strategy of Player 1, Player 2 can guarantee payoff arbitrarily close to  $W(x, y)$  by deviating to a pure control. Consequently, every Nash equilibrium of the original game gives Player 2 payoff at least  $W(x, y)$ .*

*Proof.* For each fixed pure control  $D$ , the response induced by any admissible strategy of Player 1 is a nonnegative cumulative dividend process. By theorem 2.2, Player 2's payoff is at least  $J_2^{x, y}(D, 0)$ , pathwise in the opponent's private randomizer and hence also after averaging. Take the supremum over  $D$ .  $\square$

### 3 The time-zero concession and trigger profile

Fix throughout this section  $0 < x < y$  and set

$$z_* := \min\{x, a_0\}, \quad c_* := y - z_*. \quad (11)$$

The richer Player 2 pays  $c_*$  at time zero, thereby reducing its reserve to  $z_*$ , and then follows the classical control from that reserve:

$$D_t^* := c_* + C_t^{z_*}, \quad t \geq 0. \quad (12)$$

Because  $z_* \leq a_0$ ,  $C_0^{z_*} = 0$ , so  $D_0^* = c_*$ . The remaining reserve after the concession is strictly positive.

For a nondecreasing opponent path  $d$  with  $d(0) = c_*$ , define its residual path by

$$(d - c_*)_t := d(t) - c_*, \quad t \geq 0, \quad (d - c_*)_{0-} := 0.$$

It is again a nonnegative nondecreasing path. Define Player 1's pure strategy by

$$S^*[d] := \begin{cases} G^{x; z_*}[d - c_*], & d(0) = c_*, \\ 0, & d(0) \neq c_*. \end{cases} \quad (13)$$

Thus Player 1 invokes the known predatory strategy precisely after observing the prescribed concession; every other time-zero action is punished by permanent withholding.

**Lemma 3.1** (Admissibility and induced continuation). *The map  $S^*$  in (13) is a pure admissible nonanticipative strategy from reserve  $x$ ,  $D^*$  is a pure admissible control from reserve  $y$ , and the pair  $(S^*, D^*)$  is control-inducing. After the time-zero payment  $c_*$ , its induced continuation is exactly the equilibrium  $(G^{x; z_*}, C^{z_*})$  from theorem 1.1.*

*Proof.* The level set  $\{d : d(0) = c_*\}$  is measurable and is observable at time zero. On that set,  $d - c_*$  is a valid opponent path and  $G^{x;z_*}$  is pathwise admissible and nonanticipative; off that set, the zero control is admissible. Hence  $S^*$  is an admissible pure strategy. For  $D^*$ , the time-zero jump  $c_* < y$  leaves reserve  $z_*$ , and thereafter admissibility is exactly that of  $C^{z_*}$ . Because the second component is a control, the induced system is triangular and unique. Substitution of (12) into (13) gives  $S^*[D^*] = G^{x;z_*}[C^{z_*}]$ , proving the continuation claim.  $\square$

Define the equilibrium payoff delivered to Player 2 by

$$P(x, y) := c_* + v_0(z_*) = y - z_* + v_0(z_*). \quad (14)$$

**Theorem 3.2** (Exact regret of the trigger profile). *The pure profile  $(S^*, D^*)$  has the following properties.*

- (i) *Player 1 is an exact best response to  $D^*$ , even against arbitrary randomized strategy deviations.*
- (ii) *Player 2's equilibrium payoff is  $P(x, y)$ .*
- (iii) *The supremum of Player 2's payoff over all randomized strategy deviations is exactly  $W(x, y)$ . Hence its exact deviation gain is*

$$\varepsilon_*(x, y) := W(x, y) - P(x, y) \geq 0. \quad (15)$$

- (iv) *Consequently  $(S^*, D^*)$  is a  $(0, \varepsilon_*(x, y))$ -Nash equilibrium of the full randomized-strategy game.*

*Proof.* After the deterministic time-zero concession, Player 1 faces exactly the control  $C^{z_*}$  in the source equilibrium with reserves  $(x, z_*)$ . Every pure control deviation by Player 1 is therefore unprofitable by (5). Conditioning on the private randomizer of an arbitrary randomized deviation, exactly as in the proof of theorem 2.1, proves the first claim.

Player 2 receives the immediate dividend  $c_*$  and, as the designated control player in the imported continuation equilibrium, receives  $v_0(z_*)$  thereafter. This proves (ii).

Now let  $D \in \mathcal{D}(y)$  be a pure control deviation. If  $D_0 = c_*$ , then  $\bar{D} := D - c_*$  is an admissible control from reserve  $z_*$ , and the trigger invokes  $G^{x;z_*}$ . By (6),

$$J_2^{x,y}(D, S^*[D]) = c_* + J_{\text{ctrl}}^{x,z_*}(\bar{D}, G^{x;z_*}[\bar{D}]) \leq c_* + v_0(z_*) = P(x, y). \quad (16)$$

If  $D_0 \neq c_*$ , then  $S^*[D] = 0$  and

$$J_2^{x,y}(D, S^*[D]) = J_2^{x,y}(D, 0) \leq W(x, y). \quad (17)$$

We prove in theorem 4.1 that

$$W(x, y) \geq v_0(y) \geq P(x, y). \quad (18)$$

Equations (16)–(18) show that the supremum over pure control deviations is at most  $W(x, y)$ . Conversely, for every  $D$ , the induced response  $S^*[D]$  is a nonnegative dividend process, so theorem 2.2 gives

$$J_2^{x,y}(D, S^*[D]) \geq J_2^{x,y}(D, 0).$$

Taking suprema proves that the pure-control deviation value is exactly  $W(x, y)$ . By theorem 2.1, no randomized strategy deviation can do better; pure controls are themselves permitted deviations, so the randomized deviation value is also exactly  $W(x, y)$ . This proves (iii) and (iv).  $\square$

**Remark 3.3** (Why the target  $z_*$  is natural). Suppose one pays a concession  $y - p$  and restarts the imported equilibrium with Player 2 as the designated control player from some reserve  $p \leq x$ . Its total payoff is  $y - p + v_0(p)$ . Since  $v'_0(p) > 1$  below  $a_0$  and  $v'_0(p) = 1$  above  $a_0$ , this expression is maximized over  $0 < p \leq x$  by every  $p \in [a_0, x]$  when  $x \geq a_0$ , and uniquely by  $p = x$  when  $x < a_0$ . Thus  $z_* = \min\{x, a_0\}$  is an optimal and technically simplest choice in this concession-and-restart family. Withholding is also the harshest possible off-trigger response by theorem 2.2.

## 4 Quantitative control of the remaining error

The exact error in equation (15) is a one-player singular-control value. We now bound it using only the classical dividend verification inequalities and the Laplace transform of a Brownian hitting time.

For  $g \geq 0$ , define

$$\Delta(g) := \max_{0 \leq s \leq g} (\widehat{v}(s) - v_0(s)), \quad \Delta_\infty := \sup_{s \geq 0} (\widehat{v}(s) - v_0(s)). \quad (19)$$

Both are finite: the difference is continuous and both value functions are affine with slope one above the larger of their barriers. Also  $\Delta(g) \downarrow 0$  as  $g \downarrow 0$ .

Set

$$\kappa := \frac{\mu_0 + \sqrt{\mu_0^2 + 2r\sigma^2}}{\sigma^2}. \quad (20)$$

**Proposition 4.1** (Bounds for the punishment value). *For every  $0 < x < y$ ,*

$$v_0(y) \leq W(x, y) \leq v_0(y) + e^{-\kappa x} \Delta(y - x). \quad (21)$$

Consequently,

$$0 \leq \varepsilon_*(x, y) \leq \underbrace{v_0(y) - v_0(z_*) - (y - z_*)}_{\text{concession loss}} + e^{-\kappa x} \Delta(y - x). \quad (22)$$

In particular  $W(x, y) \geq v_0(y) \geq P(x, y)$ , as used in theorem 3.2.

*Proof.* Let  $D^0$  be the optimal classical control from reserve  $y$ . The classical verification martingale, stopped at  $\tau_X(0) \wedge \tau_Y(D^0)$ , gives

$$v_0(y) = \mathbb{E} \left[ \int_{[0, \tau_X(0) \wedge \tau_Y(D^0)]} e^{-rt} dD_t^0 + \mathbf{1}_{\{\tau_X(0) < \tau_Y(D^0)\}} e^{-r\tau_X(0)} v_0(Y_{\tau_X(0)}^{D^0}) \right].$$

Since  $\widehat{v} \geq v_0$ , using the same control in the competitive payoff yields  $J_2^{x,y}(D^0, 0) \geq v_0(y)$  and therefore the lower bound in (21).

For the upper bound, fix arbitrary  $D \in \mathcal{D}(y)$  and abbreviate

$$\tau := \tau_X(0) \wedge \tau_Y(D).$$

The classical verification supermartingale for  $v_0$  gives

$$v_0(y) \geq \mathbb{E} \left[ \int_{[0, \tau]} e^{-rt} dD_t + \mathbf{1}_{\{\tau_X(0) < \tau_Y(D)\}} e^{-r\tau_X(0)} v_0(Y_{\tau_X(0)}^D) \right]. \quad (23)$$

On the event  $\{\tau_X(0) < \tau_Y(D)\}$ , the uncontrolled reserve  $x + \mu_0 t + \sigma B_t$  reaches zero continuously, so

$$Y_{\tau_X(0)}^D = y - x - D_{\tau_X(0)} \in (0, y - x]. \quad (24)$$

Subtracting the terminal  $v_0$  term in (23) from the payoff (2), and using (24), gives

$$J_2^{x,y}(D, 0) \leq v_0(y) + \Delta(y - x) \mathbb{E} \left[ e^{-r\tau_X(0)} \mathbf{1}_{\{\tau_X(0) < \infty\}} \right]. \quad (25)$$

The standard hitting-time calculation for  $x + \mu_0 t + \sigma B_t$  yields

$$\mathbb{E} \left[ e^{-r\tau_X(0)} \mathbf{1}_{\{\tau_X(0) < \infty\}} \right] = e^{-\kappa x}, \quad (26)$$

where  $\kappa$  is the positive solution of  $\frac{\sigma^2}{2} \kappa^2 - \mu_0 \kappa - r = 0$ . Taking the supremum over  $D$  proves the upper bound in (21).

Finally,  $v'_0 \geq 1$  implies

$$v_0(y) - v_0(z_*) \geq y - z_*,$$

so  $v_0(y) \geq P(x, y)$ . Subtracting (14) from (21) gives (22).  $\square$

**Corollary 4.2** (Asymptotic exactness). *The trigger profile has the following regimes of vanishing regret.*

(i) *For every fixed  $x > 0$ ,*

$$\lim_{y \downarrow x} \varepsilon_*(x, y) = 0.$$

(ii) *If  $x \geq a_0$ , then the concession loss vanishes and*

$$0 \leq \varepsilon_*(x, y) \leq e^{-\kappa x} \Delta(y - x) \leq e^{-\kappa x} \Delta_\infty. \quad (27)$$

*Hence the error converges to zero exponentially as  $x \rightarrow \infty$ , uniformly over all  $y > x$ .*

(iii) *If  $x < a_0$ , then  $P(x, y) < v_0(y) \leq W(x, y)$ , so this particular trigger profile is never an exact equilibrium.*

(iv) *If  $x \geq a_0$ , the trigger profile is an exact reversed strategy–control Nash equilibrium if and only if  $W(x, y) = v_0(y)$ .*

*Proof.* For (i), continuity of  $v_0$  and (3) imply that the concession loss tends to zero as  $y \downarrow x$ , while  $\Delta(y - x) \rightarrow 0$ . If  $x \geq a_0$ , then  $z_* = a_0$  and the affine tail of  $v_0$  gives  $v_0(y) = y - a_0 + v_0(a_0) = P(x, y)$ , proving (ii) and (iv). If  $x < a_0$ , then  $z_* = x$  and  $v'_0(s) > 1$  on a nonempty subinterval of  $(x, y)$ , so  $v_0(y) - v_0(x) > y - x$ ; this proves (iii).  $\square$

## 5 What has and has not been solved

The original open problem asks for an *exact* equilibrium, valid for every parameter choice and every  $0 < x < y$ , in which the poorer player uses a strategy and the richer player uses a control. The results above do not prove that statement. They do establish four rigorous pieces of progress:

- (1) an explicit pure reversed-role profile for every  $0 < x < y$ ;
- (2) an exact best-response result for the poorer strategic player;
- (3) an exact formula  $W(x, y) - P(x, y)$  for the richer control player's regret, valid against the full randomized-strategy deviation class;
- (4) an explicit error bound and two regimes in which the error vanishes.

The remaining scalar quantity is the value

$$W(x, y) = \sup_{D \in \mathcal{D}(y)} J_2^{x,y}(D, 0).$$

In the coordinates  $u = x$  and  $g = y - x$ , the uncontrolled rival reserve follows  $u + \mu_0 t + \sigma B_t$ , while the lead evolves as  $g - D_t$ . Thus  $W$  is a one-player singular-control problem with a random transition to the monopoly value  $\hat{v}$  when the first coordinate reaches zero. Its formal variational inequality is

$$\max \left\{ \frac{\sigma^2}{2} U_{uu} + \mu_0 U_u - rU, 1 - U_g \right\} = 0, \quad (28)$$

with boundary data  $U(0, g) = \hat{v}(g)$  for  $g > 0$  and  $U(u, -u) = 0$ . Equation (28) is included only to identify the remaining problem; no unproved regularity or verification claim for it is used in this paper.

The security-value corollary shows why this problem is intrinsic rather than an artifact of our trigger. No strategy of the poorer player can make a fixed control of the richer player worse than facing a rival who withholds, because any dividend paid by the poorer rival can only accelerate the richer player's access to the monopoly value. Therefore every exact reversed equilibrium must give the richer player at least  $W(x, y)$ . Our construction gives it the largest payoff available within the natural concession-and-restart family, but that payoff is at most  $v_0(y)$ . Exactness can therefore occur in this family only when the punishment premium  $W(x, y) - v_0(y)$  vanishes.

## 6 Why the compactness manuscript cannot be repaired locally

For completeness, we record three independent obstructions in the submitted compactness argument. These points explain why the explicit construction above is not merely a stylistic alternative.

### 6.1 The deviation class changes when the profile is randomized

The manuscript verifies the richer player's inequality only for deviations that are controls, while its candidate strategic component is a randomized causal kernel. In the original game, both players may deviate to arbitrary randomized strategies. The reduction from strategy deviations to control deviations in [1, Lemma 3.7(iii)] applies when *both equilibrium components are pure strategies*. A deviating strategy can react to the observed opponent dividend path and may thereby infer information about the opponent's private randomizer; its law is not reproduced by an opponent-independent control. The proposed theorem therefore proves a best-response condition for a smaller game. Our construction avoids this problem by being pure and then applying theorem 2.1.

### 6.2 On-path tightness does not give off-path convergence

A strategy must specify a response after every possible deviation path, not merely almost surely under the equilibrium law. There is no fixed reference probability on the opponent-path coordinate. The following elementary example isolates the failure. Let the opponent signal be  $d \in [0, 1]$  and let a response kernel be

$$K_n(\cdot \mid d) = \delta_{b_n(d)},$$

where  $b_n(d)$  is the  $n$ th binary digit of  $d$  (with any fixed convention at dyadic points). At the on-path signal  $d = 0$ , all kernels agree. Yet for every subsequence  $n_k$  one can choose a non-dyadic  $d$  whose digits at positions  $n_k$  alternate between zero and one; then  $K_{n_k}(\cdot \mid d)$  does not converge. Hence



no subsequence is simultaneously convergent after all deviations. The example can be embedded in bounded-rate control paths by encoding  $d$  during an early time interval and responding later. Equilibrium-law tightness cannot supply the missing off-path compactness.

### 6.3 Bounded-rate controls are not payoff-dense at time zero

Take controls that pay the entire reserves at time zero:

$$L_0 = x, \quad D_0 = y.$$

Both firms default simultaneously, so Player 2 receives its immediate dividend  $y$  and no continuation value. Now keep  $L$  fixed and approximate the time-zero jump of  $D$  by absolutely continuous bounded-rate controls  $D^n$  with  $D_0^n = 0$ . Player 1 still defaults at time zero, while Player 2 has reserve  $y$  and therefore immediately receives the monopoly continuation value  $\widehat{v}(y)$ . Thus

$$J_2^{x,y}(D^n, L) = \widehat{v}(y) > y = J_2^{x,y}(D, L) \quad \text{for every } n.$$

The strict inequality follows from  $\widehat{v}(0) = 0$ ,  $\widehat{v}' \geq 1$ , and  $\widehat{v}' > 1$  below the positive monopoly barrier. Consequently, the proposed delay-and-convolution approximation is not payoff-dense. Any graph topology broad enough to regard the delayed payments as approximating the time-zero atom still cannot make the payoff functional continuous at this example. The singular-limit step in the manuscript is therefore false even before the off-path issue is considered.

## 7 Conclusion

The exact reversed strategy-control equilibrium problem remains unresolved in full generality. The compactness route in the submitted manuscript fails for structural, not cosmetic, reasons. The trigger construction developed here gives a rigorous and quantitative substitute: the poorer player is exactly optimal, the richer player's full deviation value is identified with the one-player punishment value  $W(x, y)$ , and the resulting regret is explicitly controlled and asymptotically vanishing in two natural regimes.

A complete solution now has a sharply defined target. One must either solve the one-player problem  $W$  and show that  $W = v_0(y)$  in the relevant states, or design a different continuation that grants the richer control player at least its unavoidable security value  $W(x, y)$  while preserving the poorer strategic player's best-response property. Nothing in the present argument proves that such an exact construction is impossible.

## References

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